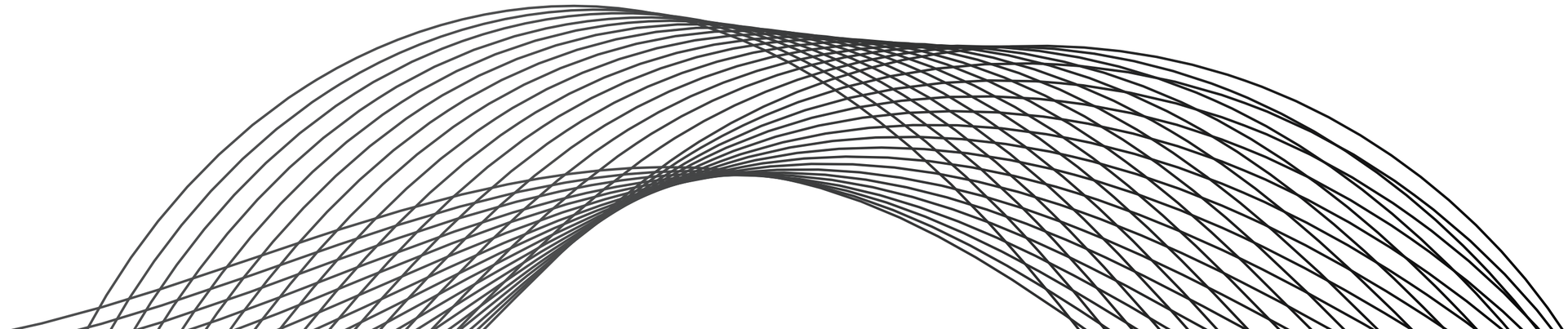
QR CODE GENERATION & SCANNING MICROSERVICE

GROUP 7



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WORK TEAM

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TECHNOLOGY OVERVIEW

The Core Problem:
Heavy image-processing and cryptographic logic often bloat monolithic applications.

Decoupled Verification:
This microservice separates the physical-to-digital bridge (QR processing) from the main business logic.

Platform Agnosticism:
Standardizes how mobile apps, browsers, and hardware scanners interact via simple HTTP APIs.

Stateless Scalability:
Validates ticket authenticity instantly using HMAC-SHA256 cryptographic signatures rather than querying heavy SQL databases.

ARCHITECTURE & WORKFLOW

Creation

The backend packages routing identifiers, TTL constraints, or cryptographic signatures into a Verification URL, rendered as a Base64 QR code.

Execution & Feedback Loop

The API serves stateless canonical data, or mutates state in-memory (e.g., ticket burning). Display devices poll via JavaScript for real-time UI updates.

Verification & Scanning

Scanners route directly to the API Gateway. The microservice validates device fingerprints, payload integrity, or simply routes stateless requests.

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HANDS-ON DEMONSTRATION

Live Demo 1:
Event Check-In: Demonstrating stateful broadcast verification with a live attendee counter and Time-to-Live (TTL) session limits.

Live Demo 2:
Digital Tickets: Scanning a single-use cryptographic ticket to demonstrate forgery prevention and state mutation (redeemed status).

Live Demo 3:
Product Catalog: Scanning a stateless, multi-use canonical route to prove infinite-scan behavior for physical assets.

Source Code:
https://github.com/DawnRighteous21/SOA-QRCode-Microservice

MICROSERVICE INTEGRATION

Language-Agnostic Utility Service: The QR API acts as an independent, headless node that communicates via standard REST/JSON HTTP protocols, meaning it can be consumed by any frontend or backend technology.

Decoupled Generation: The main application delegates the heavy lifting (image processing and cryptographic signing) to the microservice. The microservice returns a lightweight JSON payload containing a Base64 image string and a verification URL.

Direct Verification Routing: Physical scanners (like mobile phones) do not route traffic through the main application. They hit the QR microservice directly, keeping validation traffic and state mutation isolated from core business servers.

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APPLICABLE SEMESTER PROJECTS

- **E-Commerce & Retail Management:**
  Utilizing the "Product QR" endpoints to attach multi-use canonical routing tags to physical inventory.

- **University Campus Systems:**
  Managing club event check-ins using the broadcast QR feature.

- **Cinema & Transportation Booking:**
  Issuing secure, single-use digital tickets that automatically invalidate themselves in the database upon entry.

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BEST PRACTICES & LIMITATIONS

BP

Optimize Error Correction (ECC):
Configure data recovery levels (7% to 30%) based on the physical deployment environment to ensure readability if damaged.

BP

Dynamic Over Static Routing:
Embed routing URLs rather than hard-coding raw data. This allows backend updates without reprinting the physical code.

BP

Minimize Payload Density:
QR matrix complexity is directly proportional to its data volume. Minimizing the payload reduces the module count, which accelerates hardware decoding algorithms and lowers scan failure rates.

L

Inherent Security Vulnerabilities:
Standard QR codes are public visual data, making them entirely defenseless against unauthorized duplication, screenshots, or photocopying.

L

Hardware & Environmental Dependency:
Scanning success relies strictly on the end-user's camera resolution, ambient lighting, and the presence of a mandatory "Quiet Zone" (white margin).

APPLYING THE BEST PRACTICES

Dynamic & Lightweight Payloads:

Our microservice acts as a centralized routing hub. QR codes store only lightweight verification URLs rather than hard-coded data.

Mitigating Physical Vulnerabilities:

Counters the risk of unauthorized QR code duplication (like screenshots) by enforcing stateful, single-use ticket tracking at the API level.

Cryptographic Tamper-Proofing:

Secures ticket data using server-side HMAC-SHA256 signatures, ensuring authenticity and preventing users from forging or altering payloads.

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REUSABLE SOURCE CODE & DOCUMENTATION

- **Ready for Adoption:**
  The microservice is fully open-source.

- **Interactive Testing:**
  Clone the repository, run **dotnet run**, and immediately test the API using the built-in Swagger UI at http://localhost:5077/swagger.

- **Documentation:**
  Comprehensive architecture details and integration strategies are documented in the repository **README.md**

- **Visual Dashboard:**
  Includes a plug-and-play HTML frontend for instant local demonstration.

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